

The image shows a Visual Studio Code editor with a Python script named `Cancer.py` open. The script uses `pandas` to read `cancer_data.csv`, `sklearn.linear_model` for `LogisticRegression`, and `sklearn.preprocessing` for `LabelEncoder`. It also uses `joblib` to save the trained model as `model.pkl`. The terminal window at the bottom shows the execution output, including a convergence warning and the model's coefficients.

```
1 import pandas as pd
2 from sklearn.linear_model import LogisticRegression
3 from sklearn.preprocessing import LabelEncoder
4 import joblib
5
6 data = pd.read_csv("cancer_data.csv")
7 labelencoder = LabelEncoder()
8 data['diagnosis_encoded'] = labelencoder.fit_transform(data['diagnosis'])
9 x= data[["radius_mean", "texture_mean", "perimeter_mean", "area_mean", "smoothness_mean", "compactness_mean", "concavity_mean", "concave points_mean", "symmetry_mean", "fractal_dimension"]]
10 y= data['diagnosis_encoded']
11
12 model = LogisticRegression()
13 model.fit(x,y)
14 print(" coef =",model.coef_)
15 print("intercept =",model.intercept_)
16
17 joblib.dump(model, 'model.pkl')
18 print("Model saved as model.pkl ")
```

Terminal Output:

```
PS D:\coding> & c:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/Cancer.py
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\linear_model\_logistic.py:406: ConvergenceWarning: lbfgs failed to converge
after 100 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
coef = [[-0.82602533 -0.39068411 -0.36582212  0.01686076  0.02930868  0.1406838
  0.19725844  0.08284938  0.04070186  0.00802399 -0.03490221 -0.31465127
 -0.09847498  0.10514417  0.00268178  0.03053159  0.04240827  0.01084177
  0.00985935  0.00281148 -0.07558153  0.45080564  0.294833  0.01244202
  0.05352652  0.44098701  0.54884048  0.15995218  0.13017132  0.04200612]]
intercept = [-0.15245352]
Model saved as model.pkl
PS D:\coding>
```

File Edit Selection View Go Run Terminal Help

coding

Update

EXPLORER

CODING

- New folder
- aj.py
- Assignment 2_PowerOutput.py
- cancer_data.csv
- Cancer_predictor.py
- Cancer.py
- data.csv
- KNN_Dataset.csv
- KNN_FuelConsumption.py
- LoadExtension_ALPHIN.py
- LoadExtension.py
- machine_failure.csv
- Machine_Failure.py
- ml_program.py
- model.pkl
- power_data.csv
- sasffds.py

Cancer_predictor.py > ...

```
1 import joblib
2 model = joblib.load("model.pkl")
3 input_data = [[1,323,424,546,646,646,78,979,800,900,100,200,300,400,500,600,700,800,900,1000,1100,1200,1300,1400,1500,1600,1700,1800,1900,2000]]
4 prediction = model.predict(input_data)
5 print("Prediction for the input data:", prediction)
6 if prediction[0] == 1:
7     print("The predicted diagnosis is: Malignant")
8 else:
9     print("The predicted diagnosis is: Benign")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python Python

Increase the number of iterations to improve the convergence (max_iter=100).
You might also want to scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
coef = [[-0.82602533 -0.39068411 -0.36582212 0.01686076 0.02930868 0.1406838
0.19725844 0.08284938 0.04070186 0.00802399 -0.03490221 -0.31465127
-0.09847498 0.10514417 0.00268178 0.03053159 0.04240827 0.01084177
0.00985935 0.00281148 -0.87558153 0.45080564 0.294833 0.01244202
0.05352652 0.44098701 0.54884048 0.15995218 0.13017132 0.04200612]]
intercept = [-0.15245352]
Model saved as model.pkl
PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/Cancer_predictor.py
C:\Users\alphi\AppData\Local\Programs\Python\Python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names
, but LogisticRegression was fitted with feature names
warnings.warn(
Prediction for the input data: [1]
The predicted diagnosis is: Malignant
PS D:\coding>
```

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